

01 - Backend Optimisation: List & Search

Imagine the B2B SaaS Dashboard from hello again. Businesses use it as a CRM, communication/newsletter tool and also manage their loyalty assets. One main use case of such a CRM is listing, searching & filtering for contacts.

Take the following data model, consisting of 3 tables as given:

Unset

AppUser

- id
- first_name
- last_name
- gender
- customer_id
- phone_number
- created
- address_id (FK)
- birthday
- last_updated

Address

- id
- street
- street_number
- city_code
- city
- country

CustomerRelationship

- id
- appuser_id (FK)
- points
- created
- last_activity

Task

- Set up a Django project that reflects that data structure
- Write a script to insert ~3 Mio data entries (AppUser, Address and a Customer Relationship) - you can use random values
- For the Database feel free to use your personal preference (Sqlite, Postgresql, etc.)
- Implement a View for the data structure that lists your entries. It should join all 3 tables, since the response should include attributes from all 3 tables
- The view should be able to sort, filter and list by any field
- The view should also come with pagination support
- Benchmark your view with certain queries (measure how long it takes to return results)
 - E.g. Filter by a name
 - Sort by attribute
 - Load initial list including pagination
- Now think about performance optimisations, implement them and compare them with your initial benchmarks